

***FEDERAL URDU UNIVERSITY OF ARTS, SCIENCE
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*JIRZO – AI-Based Behavioral Risk Assessment
for Adaptive Authentication*

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JIRZO - AI-Based Behavioral Risk Assessment for Adaptive Authentication

Problem Identification

The initial investigation for JIRZO began with examining how traditional authentication systems determine whether a user should be allowed to access an account.

In a traditional authentication system, a user provides credentials, usually a username and password. The system verifies these credentials against the stored account information. If the credentials are correct, the authentication process is considered successful and the user is allowed to log in.

The problem with this approach is that **valid credentials do not necessarily prove that the person attempting the login is the legitimate account owner.**

For example, if a user's username and password have been obtained by another person through credential theft or phishing, the credentials provided to the authentication system are still correct. The system may therefore consider the authentication successful even though the person attempting the login is not the legitimate user.

This creates a gap between **credential validity** and **user authenticity**.

The main question that emerged from this problem is:

If the credentials provided during a login attempt are correct, how can we determine whether the person attempting the login is actually the legitimate user?

To investigate this problem, I considered whether information associated with the login attempt could provide additional evidence about the user's identity and behavior.

A login attempt contains more information than just the username and password. For example, JIRZO can potentially collect information such as:

- IP address
- Region
- Login time
- Browser
- Whether the device has previously been used by the user
- Failed password attempts
- OTP requests

These characteristics can be used to examine whether the current login attempt is consistent with the user's previous login behavior.

However, these characteristics should not be treated individually as proof of unauthorized access. A new IP address, different region, unfamiliar device, or unusual login time can also occur during legitimate use.

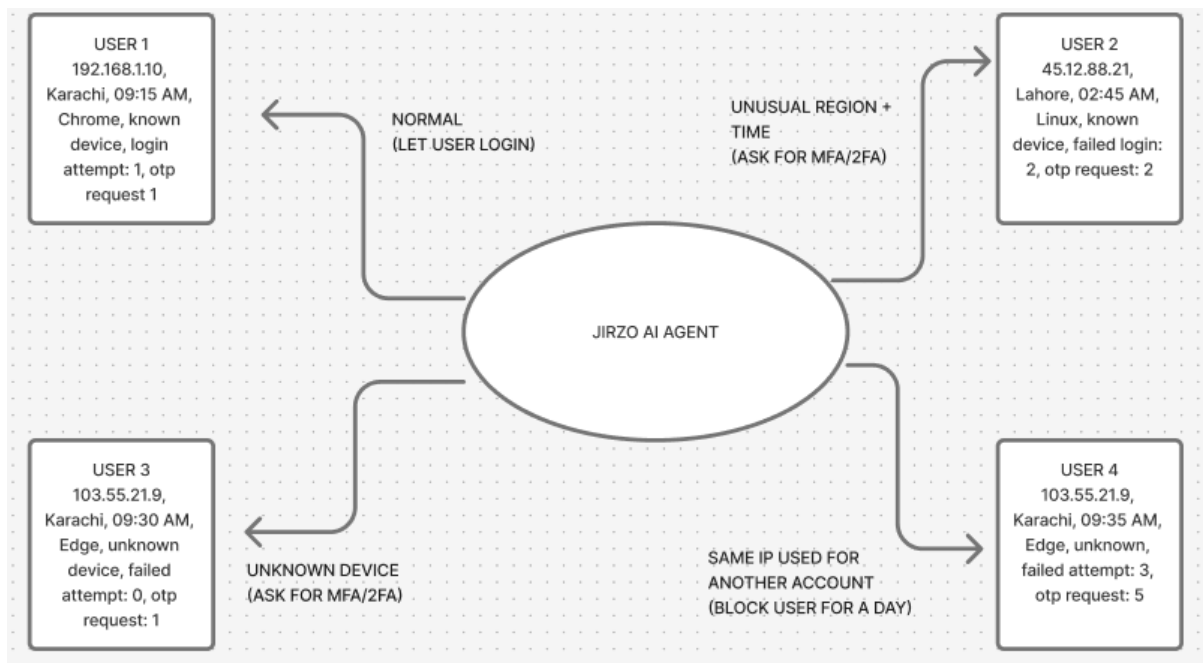
Therefore, the investigation focuses on the possibility of **combining multiple login-related signals and comparing them with previous user behavior** rather than relying on a single characteristic.

Problem Formulation

Traditional authentication systems primarily determine whether a user should be granted access by verifying the correctness of their credentials. However, the correctness of credentials alone does not necessarily confirm that the person attempting the login is the legitimate account owner, particularly when valid credentials have been compromised.

Therefore, the problem is to investigate whether additional information associated with a login attempt can be used to assess the legitimacy of the authentication attempt. This includes examining behavioral and contextual signals such as the user's IP address, region, login time, browser, device, previous device usage, previous login behavior, failed login attempts, and OTP requests.

The proposed investigation focuses on analyzing these signals collectively and comparing the current login attempt with the user's previous behavior to determine whether the authentication attempt appears consistent with normal behavior or shows unusual behavioral patterns.



Dataset for AI Agent:

Log_id	user	Ip_address	region	login_time	Browser	Device	Failed Attempt	OTP Req	decision
L001	01	192.168.1.10	Karachi	09:15 AM	Chrome	known	0	1	Normal
L002	02	192.168.1.11	Karachi	10:20 AM	Chrome	known	2	1	Normal
L003	04	45.12.88.21	Lahore	02:45 AM	Linux	known	2	2	UNUSUAL REGION + TIME
L004	02	103.55.21.9	Karachi	09:30 AM	Edge	unknown	0	1	UNKNOWN DEVICE
L005	06	103.55.21.9	Karachi	09:32 AM	Edge	unknown	3	4	Multiple failed attempts
L006	09	103.55.21.9	Karachi	09:35 AM	Edge	unknown	2	3	SAME IP USED FOR ANOTHER ACCOUNT

EVALUATION IN THE CONTEXT OF PEAS:

P – Performance Measure: Response Time, Anomaly Detection Rate, Accuracy, Efficiency

E – Environment: The machine where the system is operating

A – Actuators: If user marked as **suspicious** it will ask for MFA or 2FA or may block the user.
If user marked as **normal** it will let user login

S – Sensors: User ID, IP Address, Region, Login Time, Browser, Device, Failed Attempts, OTP Request

Environmental Properties:

1. **Fully Observable:** Jirzo is fully observable because it analyzes the user's whole behavior during login.
2. **Stochastic:** Jirzo's decision is based on previous events, and it will include probability and sometimes randomness.
3. **Sequential:** Jirzo makes decisions based on previous login event.

4. **Static Environment:** Jirzo makes its decision on a fixed login event and doesn't modify the environment during that decision
5. **Discrete:** Jirzo deals with points (a finite number) based on every attribute from the login event.
6. **Single Agent:** Jirzo is a single agent because it makes decisions independently.